



Dear Representative of the Dresden City Council,

I would also like to briefly introduce another project of Khmelnytskyi National University (KhNU), funded by the **Ministry of Education and Science of Ukraine**, which addresses one of the key environmental challenges relevant for modern cities — **polymer waste management**.

The project focuses on the **development of a smart thermochemical system for converting polymer waste into liquid fuel, supported by an energy-autonomous solar power supply**.

From a municipal perspective, the project **explores decentralised approaches to waste processing**, which may reduce dependence on large-scale centralised infrastructure and contribute to local energy resilience.

The core of the project lies in:

- optimisation of thermochemical conversion processes to maximise useful fuel output while minimising harmful by-products
- development of an **automated control system** that adapts the process to different types of waste and available solar energy
- integration of **real-time environmental monitoring**, including emissions control and fuel quality assessment
- utilisation of by-products in line with **circular economy principles**

One of the important outcomes of the project is the creation of a **decision-support framework for local authorities**, enabling them to assess the feasibility of implementing decentralised waste-to-energy solutions at the community level.

We see particular relevance of this work for cities that are:

- exploring sustainable waste management solutions
- aiming to increase energy autonomy
- interested in circular economy approaches

In the context of our future visit to Dresden, we would be very interested in discussing:

- existing municipal approaches to waste processing and circular economy
- perspectives on decentralised waste-to-energy solutions
- possible areas for knowledge exchange or pilot cooperation

We believe that this topic may offer an additional dimension for collaboration between our institutions and cities.

Yours sincerely,
Kateryna Skyba